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Two Spanning Disjoint Paths with Required length in Generalized Hypercubes

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Abstract

This work investigates 2RP-property of a generalized hypercube G. Given any four distinct vertices u, v, x and y in G, let l_1 and l_2 be two integers such that l_1 (l_2) is not less than the distance between u and v (x and y), and l_1+l_2 is equal to the number of vertices in G minus two. Then, there exist two vertex-disjoint paths P_1 and P_2 such that (1) P_1 is a path joining u and v with length of l_1; (2) P_2 is a path joining x and y with length of l_2, and (3) $P_1 \cup P_2$ spans G except some special conditions. This work shows that a $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies 2RP-property, where $m_i \geq 4$ for all $1 \leq i \leq r$.

1 Introduction

The topological structure of a multicomputer system can be modeled by an interconnection network (or a graph), the interconnection network plays an important role in some issues such as communication performance [3], hardware cost [1], embedding and fault tolerant capabilities [16], and those are all driven by a mathematical model. Paths are suitable for designing simple algorithms with low communication costs. The path embedding problem is a very important issue for a network and is widely discussed in many researches [2], [5], [6], [12], [13], [14]. Generally, an interconnection network can be modeled by a graph, and edges (vertices) in the graph represent links (vertices) in the interconnection network.

An interconnection network is usually represented by an undirected simple graph $G = (V, E)$. $V(G)$ ($E(G)$) denotes the vertex (edge) set of G, where $V(G)$ ($E(G)$) is a finite set. An edge between two vertices u and v in G is denoted by (u, v). For two distinct vertices $u, v \in V(G)$, u and v are adjacent if $(u, v) \in E(G)$. A path in G is a sequence of edges that connect adjacent vertices. A *Hamiltonian path* is a path that contains every vertex of G exactly once. A graph G is *Hamiltonian-connected* if every two distinct vertices of G are connected by a Hamiltonian path. A graph G is *k-vertex fault-tolerant Hamiltonian-connected* (*k-Hamiltonian-connected* for short) if it remains Hamiltonian-connected after removing no more than k vertices from G.

The interconnection network considered in this work is the generalized hypercube which has excellent topological properties, such as logarithmic diameter [9], vertex-symmetry [9], edge-symmetry [9], efficient communication [7], [9], [17] and high degree of fault tolerance [16]. Some basic properties of a generalized hypercube, such as diameter, wide diameter, and fault diameter have been determined in [4].

In [10], Lee et al. introduced an interesting property, called 2RP-property, as described below. Let $l(P)$ denote the length of a path P, i.e. the number of edges which connect adjacent vertices in P. The distance between vertices u and v in G, denoted by $d_G(u, v)$, is the minimum $l(P)$ for every P joining u and v. Given any four distinct vertices u, v, x, and y in a graph G, let l_1 and l_2 be two integers such that $l_1 \geq d_G(u, v)$, $l_2 \geq d_G(x, y)$, and $l_1+l_2 = |V(G)|-2$. Then, there exist two vertex-disjoint (disjoint for short) paths P_1 and P_2 such that (1) P_1 is a path joining u and v with $l(P_1) = l_1$; (2) P_2 is a path joining x and y with $l(P_2) = l_2$, and (3) $P_1 \cup P_2$ spans G. Two paths are *vertex-disjoint* if they do not share any common vertex. The 2RP-property have been studied on various graphs, such as Hypercubes [10], augmented cubes [11], arrangement graphs [15]. Since a generalized hypercube is a generalization

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of a hypercube, this work further shows that a generalized hypercube satisfies 2RP-property.

The rest of this paper is organized as follows. Section 2 formally introduces the definition and some properties of a generalized hypercube. Section 3 defines the 2RP-property for the generalized hypercube and proves that a generalized hypercube satisfies the 2RP-property. Conclusions are finally drawn in Section 4.

2 Background and Notations

This section discusses the structure and some properties of a generalized hypercube. Some notations frequently used in this work is also presented.

2.1 Generalized Hypercube graphs

Generalized hypercubes are the generalization of hypercubes [1] which were proposed for building massively parallel computer systems. Let $G(m_r, m_{r-1}, \dots, m_1)$ denote a r-dimensional generalized hypercube of order $\Pi_{i=1}^{r}(m_i)$, where $r\geq 1$ and $m_i \geq 2$ for all $1\leq i\leq r$. In other words, there are $m_r\times m_{r-1}\times\dots\times m_1$ vertices in a $G(m_r, m_{r-1}, \dots, m_1)$ and each vertex is assigned a r-digit identifier $x_r x_{r-1}\dots x_1$, where $x_i\in[0, m_i-1]$ for all $1\leq i\leq r$. In a $G(m_r, m_{r-1}, \dots, m_1)$, two vertices are adjacent if and only if their identifiers differ at exactly one digit position. Two adjacent vertices whose identifiers differ at i position are connected by an i-edge and they are i-nerighbors, where $1\leq i\leq r$. Each vertex in a $G(m_r, m_{r-1}, \dots, m_1)$ has degree $\Sigma_{i=1}^{r}(m_i-1)$.

There exist m_r supernodes $G^r[j]$ for $0\leq j\leq m_r-1$ with the vertex sets $V(G^r[j]) = \{jx_{r-1}\dots x_1 \mid 0\leq x_i\leq m_i-1$ for $1\leq i\leq r-1\}$ that form a partition of $V(G(m_r, m_{r-1}, \dots, m_1))$. A $G(m_r, m_{r-1}, \dots, m_1)$ is a complete graph of m_r $G^r[j]$s. Let N^r denote the order of a $G(m_{r-1}, m_{r-2}, \dots, m_1)$. Obviously, $|V(G^r[j])| = N^r$ for all $0\leq j\leq m_r-1$. Let $x = x_r x_{r-1}\dots x_1$ be a vertex in a $G(m_r, m_{r-1}, \dots, m_1)$, $\sigma^i(x) = x_i$ for $1\leq i\leq r$. Also, let x^j in $G^r[j]$ be the r-neighbor of x in $G^r[x_r]$ for $0\leq j\leq m_r-1$ and $j\neq x_r$. Notably, the order of dimensions of a $G(m_r, m_{r-1}, \dots, m_1)$ can be exchanged without changing its structure. For instance, a $G(4, 3, 2)$ is isomorphic to a $G(2, 3, 4)$. Figure 1 shows the structure of a $G(4, 3, 2)$. Let $N_G(u)$ denote the set of vertices adjacent to vertex u in G. For convenience, let $\langle u, u_0, u_1, \dots, u_m, v\rangle$ denote a path joining u and v, where all the vertices u, u_1, …, u_m, and v are distinct. Additionally, $\langle u, u_0, u_1, \dots, u_m, v\rangle=\langle u, P, v\rangle$ if $P = \langle u, u_0, u_1, \dots, u_m, v\rangle$.

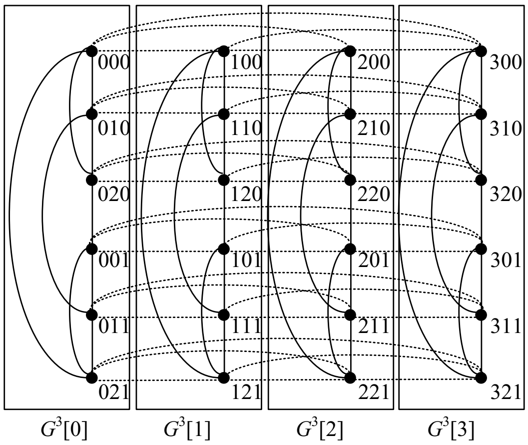

Figure 1. The structure of a $G(4, 3, 2)$.

2.2 Some properties of Generalized Hypercube graphs

This section addresses some properties of a generalized hypercube, and proves that it is a 1-Hamiltonian-connected graph.

Lemma 1 [8]. A $G(m_r, m_{r-1}, \dots, m_1)$ is Hamiltonian-connected, except m_i=2 for all $1\leq i\leq r$.

Lemma 2. A $G(m_r, m_{r-1}, \dots, m_1)$ is 1-Hamiltonian-connected, where $m_i\geq 3$ for all $1\leq i\leq r$.

Proof. This lemma is proved by induction on dimension r. For r=1, $G(m_1)$ is a complete graph of m_1 vertices and the lemma clearly holds. Assume the lemma holds for any $G(m_r, m_{r-1}, \dots, m_1)$ with $r\leq k$.

Let f be the faulty vertex in a $G(m_{k+1}, m_k, \dots, m_1)$, and $u, v \in V(G(m_{k+1}, m_k, \dots, m_1))-\{f\}$ be two distinct vertices. Since $u\neq v$, there exists a digit l such that $\sigma^l(u)\neq\sigma^l(v)$, where $1\leq l\leq k+1$. Without loss of generality, assume that $l = k+1$. Since $\sigma^{k+1}(u)\neq \sigma^{k+1}(v)$, we may assume that $u\in V(G^{k+1}[0])$ and $v\in V(G^{k+1}[1])$ without loss of generality. There are two scenarios to be considered in the following.

Case 1. ($\sigma^{k+1}(f) \notin \{\sigma^{k+1}(u), \sigma^{k+1}(v)\}$): Without loss of generality, assume that $\sigma^{k+1}(f) = 2$, and thus f is included in $G^{k+1}[2]$.

Case 1.1. ($N^{k+1}\geq 4$): Since $|V(G^{k+1}[j])|\geq 4$ for all $1\leq j\leq m_{k+1}$, there exists p^2 in $G^{k+1}[2]\backslash\{f\}$ such that $p^0\neq u$, and q^2 in $G^{k+1}[2]\backslash\{f, p^2\}$ such that $q^1\neq v$. By the induction hypothesis, there exists a Hamiltonian path H_1 in $G^{k+1}[2]\backslash\{f\}$ joining p^2 and q^2. By Lemma 1, there exists two Hamiltonian paths H_2 in $G^{k+1}[0]$ and H_3 in $G(m_{k+1}, m_k, \dots, m_1)-G^{k+1}[0]-G^{k+1}[2]$ such that H_2 joins u and p^0, and H_3 joins q^1 and v. Obviously, $\langle u, H_2, p^0, p^2, H_1, q^2, q^1, H_3, v\rangle$ is a Hamiltonian path in $G(m_{k+1}, m_k, \dots, m_1)\backslash\{f\}$. The Hamiltonian path construction of Case 1.1 is illustrated in Figure 2.

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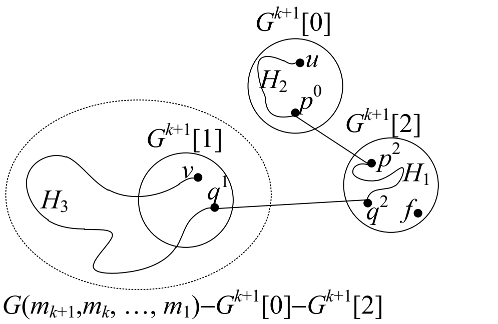
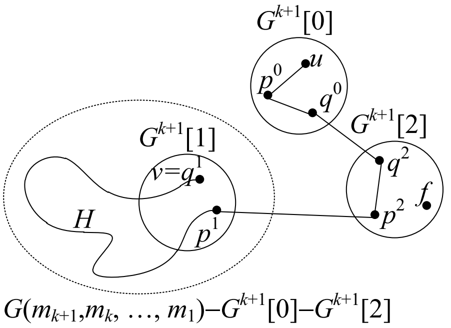
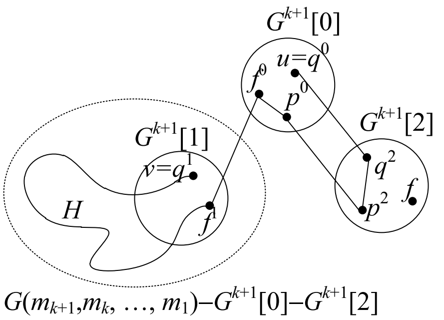


Figure 2. The Hamiltonian path constructed by Case 1.1.

Case 1.2. ($N^{k+1}=3$): There exists a vertex p^2 in $G^{k+1}[2]\backslash\{f\}$ such that $p^0\neq u$ and q^2 is included in $G^{k+1}[2]\backslash\{f, p^2\}$. The discussions depend on the locations of q^1 and q^0. When $q^1\neq v$, the proof is the same as that in Case 1.1. When $q^1=v$ and $q^0\neq u$, by Lemma 1, there exists a Hamiltonian path H in $G(m_{k+1}, m_k, \dots, m_1)–G^{k+1}[0]–G^{k+1}[2]$ joining p^1 and v. Hence, $\langle u, p^0, q^0, q^2, p^2, p^1, H, v\rangle$ is the Hamiltonian path in $G(m_{k+1},m_k, \dots, m_1)\backslash\{f\}$ as shown in Figure 3(a). When $q^1=v$ and $q^0=u$, by Lemma 1, there exists a Hamiltonian path H in $G(m_{k+1},m_k, \dots, m_1)–G^{k+1}[0]–G^{k+1}[2]$ joining f^1 and v. Hence, $\langle u, q^2, p^2, p^0, f^0, f^1, H, v\rangle$ is the Hamiltonian path in $G(m_{k+1},m_k, \dots, m_1)\backslash\{f\}$ as depicted in Figure 3(b).



(a)



(b)

Figure 3. Hamiltonian paths constructed by Case 1.2. (a) $q^1=v$ and $q^0\neq u$. (b) $q^1=v$ and $q^0=u$.

Case 2. ($\sigma^{k+1}(f)\in\{\sigma^{k+1}(u), \sigma^{k+1}(v)\}$): Without loss generality, assume that $\sigma^{k+1}(f)=\sigma^{k+1}(u)$, and then assume that both f and u are included in $G^{k+1}[0]$. There exists a vertex p^0 in $G^{k+1}[0]\backslash\{f, u\}$. By Lemma 1, there exists a Hamiltonian path H_1 in $G(m_{k+1},m_k, \dots, m_1)–G^{k+1}[0]$ joining p^2 and v because $\sigma^{k+1}(p^2)\neq\sigma^{k+1}(v)$. By the induction hypothesis, there exist a Hamiltonian path H_2 in $G^{k+1}[0]\backslash\{f\}$ joining u and p^0. Obviously, $\langle u, H_2, p^0, p^2, H_1, v\rangle$ is the Hamiltonian path in $G(m_{k+1},m_k, \dots, m_1)\backslash\{f\}$. ■

Lemma 3. Let P be a path in a $G(m_r, m_{r–1}, \dots, m_1)$. If $l(P)\geq m_r$, then P contains an m-edge where $m\neq r$.

Proof. Let $P = \langle u^0, u^1, \dots, u^k\rangle$ be a path in the $G(m_r, m_{r–1}, \dots, m_1)$, where $k\geq m_r$, and any two vertices of P is connected by r-edges. In other words, $\sigma^r(u^i)\neq\sigma^r(u^j)$ where $0\leq i, j\leq k$ and $i\neq j$. Thus, there should be $k+1>m_r$ distinct symbols. It is a contradiction because there are only m_r distinct symbols in r dimension. Therefore, this lemma follows. ■

Lemma 4. Let P be a path from u to v, and w and x be any two distinct vertices in a $G(m_r, m_{r–1}, \dots, m_1)$, where $r\geq 2$ and $m_i=4$ for all $1\leq i\leq r$. If $l(P)\geq 4^r–3$, then there exist two adjacent vertices y and z in P, such that $N_{G(m_r, m_{r–1}, \dots, m_1)}(w)\cap N_{G(m_r, m_{r–1}, \dots, m_1)}(x) \neq \{y, z\}$ and $\{w, x\}\cap\{y, z\} = \varnothing$.

Proof. Since $|N_{G(m_r, m_{r–1}, \dots, m_1)}(w)|=3r$, $|V(P)–N_{G(m_r, m_{r–1}, \dots, m_1)}(w)–\{u, v, w, x\}| \geq (4^r–2)–3r–4 \geq 1$. Thus, there exist a vertex y in $V(P)–N_{G(m_r, m_{r–1}, \dots, m_1)}(w)–\{u, v , w, x\}$. Because $N_P(y)=2$ and $y\notin N_{G(m_r, m_{r–1}, \dots, m_1)}(w)$, there exists a vertex $z\in N_P(y)$ in P such that $z\notin\{w, x\}$. Therefore, there exist two adjacent vertices y and z in P such that $N_{G(m_r, m_{r–1}, \dots, m_1)}(w)\cap N_{G(m_r, m_{r–1}, \dots, m_1)}(x) \neq \{y, z\}$ and $\{w, x\}\cap\{y, z\} = \varnothing$. ■

3 2RP-property

This section demonstrates that a $G(m_r, m_{r–1}, \dots, m_1)$ satisfies the 2RP-property, when $m_i\geq 4$ for all $1\leq i\leq r$. Let u, v, x and y be any four distinct vertices of a $G(m_r, m_{r–1}, \dots, m_1)$, and l_1 and l_2 be two integers with $l_1 \geq d_{G(m_r, m_{r–1}, \dots, m_1)}(u, v)$, $l_2 \geq d_{G(m_r, m_{r–1}, \dots, m_1)}(x, y)$ and $l_1+l_2 = |V(G(m_r, m_{r–1}, \dots, m_1))|–2$. Then, there exist two disjoint paths P_1 and P_2 such that (1) P_1 is a path joining u and v with $l(P_1) = l_1$; (2) P_2 is a path joining x and y with $l(P_2) = l_2$, and (3) $P_1\cup P_2$ spans $G(m_r, m_{r–1}, \dots, m_1)$ except the following two cases: (a) $l_1=2$ or $l_1=|V(G(m_r, m_{r–1}, \dots, m_1))|–4$ with $d_{G(m_r, m_{r–1}, \dots, m_1)}(u, v)=1$ such that $N_{G(m_r, m_{r–1}, \dots, m_1)}(u)\cap N_{G(m_r, m_{r–1}, \dots, m_1)}(v) = \{x, y\}$; (b) $l_1=2$ or $l_1=|V(G(m_r, m_{r–1}, \dots, m_1))|–4$ with $d_{G(m_r, m_{r–1}, \dots, m_1)}(u, v) = 2$ such that $N_{G(m_r, m_{r–1}, \dots, m_1)}(u)\cap N_{G(m_r, m_{r–1}, \dots, m_1)}(v) = \{x, y\}$.

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Lemma 5. A $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies 2RP-property, where $r\geq 2$ and $m_i=4$ for all $2\leq i\leq r$.

Proof. This lemma is proved by induction on r. With the aids of a computer program, the lemma holds for $r=2$. Suppose the lemma holds for any $G(m_r, m_{r-1}, \ldots, m_1)$ with $r\leq k$. To prove the lemma holds, let $r=k+1$ and G denote $G(m_{k+1}, m_k, \ldots, m_1)$ for simplicity. Without loss of generality, assume that $l_1\leq l_2$, thus $l_1 \leq (4^{k+1}-2)/2 = 2(4^k)-1$. Let u, v, x and y be 4 arbitrary distinct vertices in G. Since $x\neq y$, there exists a digit l such that $\sigma^l(x)\neq\sigma^l(y)$, where $1\leq l\leq k+1$. Without loss of generality, assume $l = k+1$. Since $\sigma^{k+1}(x)\neq\sigma^{k+1}(y)$, we may assume $x\in V(G^{k+1}[0])$ and $y\in V(G^{k+1}[1])$ without loss of generality. Five cases should be considered as follows.

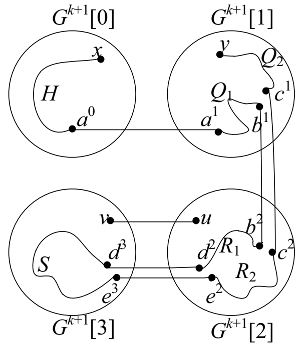
Case 1. ($\sigma^{k+1}(u)\neq\sigma^{k+1}(v)$ and $\{\sigma^{k+1}(u), \sigma^{k+1}(v)\}\cap\{\sigma^{k+1}(x), \sigma^{k+1}(y)\} = \varnothing$): Without loss of generality, assume that $\sigma^{k+1}(u)=2$, $\sigma^{k+1}(v)=3$, such that u in $G^{k+1}[2]$, v in $G^{k+1}[3]$. Three cases are necessary to be discussed depending on the values of $d_G(u, v)$ and l_1.

Case 1.1. ($d_G(u, v) \leq l_1 \leq 4^k-1$ with $d_G(u, v) = 1$):

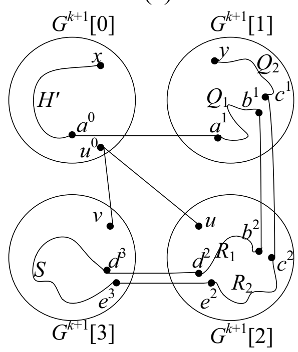
For $l_1=1$, we have $P_1 = \langle u, v\rangle$. There is a vertex a^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^1\neq y$. By Lemma 1, there are two Hamiltonian paths H in $G^{k+1}[0]$ and Q in $G^{k+1}[1]$ such that H joins x and a^0, and Q joins a^1 and y. Since $l(Q) = 4^k-1$, Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1 such that $\{b^2, c^2\}\cap\{u\} = \varnothing$. According to Lemma 2, there is a Hamiltonian path R in $G^{k+1}[2]\backslash\{u\}$ joining b^2 and c^2, and R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2. By Lemma 2, there is a Hamiltonian path S in $G^{k+1}[3]\backslash\{v\}$ joining d^3 and e^3. Hence, $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$, and P_1 and P_2 are the required paths as shown in Figure 4(a).

For $l_1=2$, only $N_G(u)\cap N_G(v) \neq \{x, y\}$ needs to be discussed. Since $N_G(u)\cap N_G(v) = \{u^0, u^1\}$, we have $\{u^0, u^1\} \neq \{x, y\}$. Without loss of generality, assume that $u^0\neq x$. Hence, $P_1 = \langle u, u^0, v\rangle$. By Lemma 2, there is a Hamiltonian path H' in $G^{k+1}[0]\backslash\{u^0\}$ joining x and a^0. Hence, $P_2 = \langle x, H', a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$, and P_1 and P_2 are the required paths as shown in Figure 4(b).

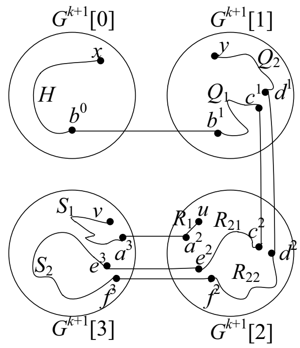
For $3 \leq l_1 \leq 4^k-1$, there are two vertices $a^2 \in N_{G^{k+1}[2]}(u)$ and b^0 in $G^{k+1}[0]\backslash\{x\}$ such that $b^1\neq y$. By Lemma 1, there are two Hamiltonian paths H in $G^{k+1}[0]$ and Q in $G^{k+1}[1]$ such that H joins x and b^0, and Q joins b^1 and y. Since $l(Q) = 4^k-1$, Q can be written as $\langle b^1, Q_1, c^1, d^1, Q_2, y\rangle$ for some vertices c^1 and d^1 such that $\{c^2, d^2\}\cap\{u, a^2\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u and a^2 with $l(R_1)=1$; (2) R_2 is a path joining c^2 and d^2 with $l(R_2)=4^k-3$, and (3) $R_1\cup R_2$ spans $G^{k+1}[2]$. By Lemma 4, R_2 can be written as $\langle c^2, R_{21}, e^2, f^2, R_{22}, d^2\rangle$ for some vertices e^2 and f^2 such that $N_{G^{k+1}[3]}(v)\cap N_{G^{k+1}[3]}(a^3) \neq \{e^3, f^3\}$ and $\{v, a^3\}\cap\{e^3, f^3\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths S_1 and S_2 such that (1) S_1 is a path joining a^3 and v with $l(S_1) = l_1-2$; (2) S_2 is a path joining e^3 and f^3 with $l(S_2) = 4^k-2-l(S_1)$, and (3) $S_1\cup S_2$ spans $G^{k+1}[3]$. Hence, $P_1 = \langle u, R_1, a^2, a^3, S_1, v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q_1, c^1, c^2, R_{21}, e^2, e^3, S_2, f^3, f^2, R_{22}, d^2, d^1, Q_2, y\rangle$ are the required paths as shown in Figure 4(c).



(a)



(b)



(c)

Figure 4. Paths P_1 and P_2 constructed by Case 1.1. (a) $l_1=1$. (b) $l_1=2$. (c) $3\leq l_1\leq 4^k-1$.

Case 1.2. ($d_G(u, v) \leq l_1 \leq 4^k-1$ with $d_G(u, v) \geq 2$):

For $d_G(u, v) \leq l_1 \leq 4^k-2$, there is a vertex

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a^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^1 \neq y$ and $a^3 \notin \{u^3, v\}$. By Lemma 1, there are two Hamiltonian paths H in $G^{k+1}[0]$ and Q in $G^{k+1}[1]$ such that H joins x and a^0, and Q joins a^1 and y. Since $l(Q)=4^k-1$, Q can be written as $\langle b^1, Q_1, c^1, d^1, Q_2, y\rangle$ for some vertices b^1 and c^1 such that $\{b^2, c^2\} \cap \{u\} = \varnothing$. By Lemma 2, there is a Hamiltonian path R in $G^{k+1}[2]\backslash\{u\}$ joining b^2 and c^2. By Lemma 4, R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2 such that $N_{G^{k+1}[3]}(v) \cap N_{G^{k+1}[3]}(u^3) \neq \{d^3, e^3\}$ and $\{v, u^3\} \cap \{d^3, e^3\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths S_1 and S_2 such that (1) S_1 is a path joining v and a^3 with $l(S_1)=l_1-1$; (2) S_2 is a path joining d^3 and e^3 with $l(S_2)=4^k-2-l(S_1)$, and (3) $S_1 \cup S_2$ spans $G^{k+1}[3]$. Hence, $P_1 = \langle u, u^3, S_1, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S_2, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$ are the required paths as shown in Figure 5(a).

For $l_1=4^k-1$, there is a Hamiltonian path S of $G^{k+1}[3]\backslash\{a^3\}$ joining u^3 and v by Lemma 2. Hence, $P_1 = \langle u, u^3, S, v\rangle$ and $P_2 = \langle x, H, a^0, a^3, a^1, Q_1, b^1, b^2, R, c^2, c^1, Q_2, y\rangle$ are the required paths as shown in Figure 5(b).

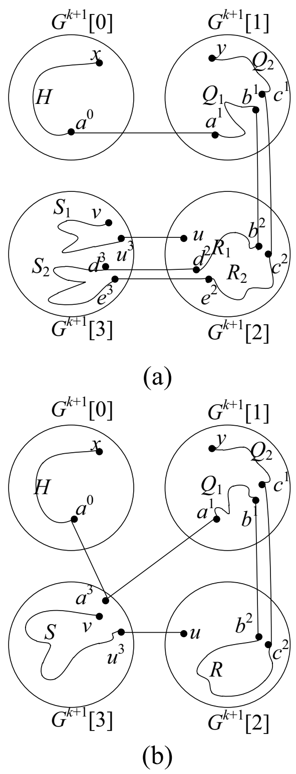


Figure 5. Paths P_1 and P_2 constructed by Case 1.2. (a) $d_G(u, v) \leq l_1 \leq 4^k-2$. (b) $l_1=4^k-1$.

Case 1.3. ($4^k \leq l_1 \leq 2(4^k)-1$):

For $l_1=4^k$, there are two vertices $a^2 \in N_{G^{k+1}[2]}(u)$ and b^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^3 \neq v$, $b^1 \neq y$, and $b^3 \notin \{v, a^3\}$. By Lemma 1, there are two Hamiltonian paths H in $G^{k+1}[0]$ and Q in $G^{k+1}[1]$ such that H joins x and b^0, and Q joins b^1 and y. Since $l(Q)=4^k-1$, Q can be written as $\langle b^1, Q_1, c^1, d^1, Q_2, y\rangle$ for some vertices c^1 and d^1 such that $\{c^2, d^2\} \cap \{u, a^2\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u and a^2 with $l(R_1)=1$; (2) R_2 is a path joining c^2 and d^2 with $l(R_2)=4^k-3$, and (3) $R_1 \cup R_2$ spans $G^{k+1}[2]$. By Lemma 2, there exists a Hamiltonian path S in $G^{k+1}[3]\backslash\{b^3\}$ joining a^3 and v. Hence, $P_1 = \langle u, R_1, a^2, a^3, S, v\rangle$ and $P_2 = \langle x, H, b^0, b^3, b^1, Q_1, c^1, c^2, R_2, d^2, d^1, Q_2, y\rangle$ are the required paths as shown in Figure 6(a).

For $4^k+1 \leq l_1 \leq 2(4^k)-3$. By Lemma 4, Q can be written as $\langle b^1, Q_1, c^1, d^1, Q_2, y\rangle$ for some vertices c^1 and d^1 such that $N_{G^{k+1}[3]}(u) \cap N_{G^{k+1}[3]}(a^2) \neq \{c^2, d^2\}$ and $\{u, a^2\} \cap \{c^2, d^2\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u and a^2 with $l(R_1)=l_1-4^k$; (2) R_2 is a path joining c^2 and d^2 with $l(R_2)=4^k-2-l(R_1)$, and (3) $R_1 \cup R_2$ spans $G^{k+1}[2]$. By Lemma 1, there exists a Hamiltonian path S' in $G^{k+1}[3]$ joining a^3 and v. Hence, $P_1 = \langle u, R_1, a^2, a^3, S', v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q_1, c^1, c^2, R_2, d^2, d^1, Q_2, y\rangle$ are the required paths as shown in Figure 6(b).

For $l_1=2(4^k)-2$, there is a Hamiltonian path R in $G^{k+1}[2]$ joining u and a^2 by Lemma 1. Hence, $P_1 = \langle u, R, a^2, a^3, S, v\rangle$ and $P_2 = \langle x, H, b^0, b^3, b^1, Q, y\rangle$ are the required paths as shown in Figure 6(c).

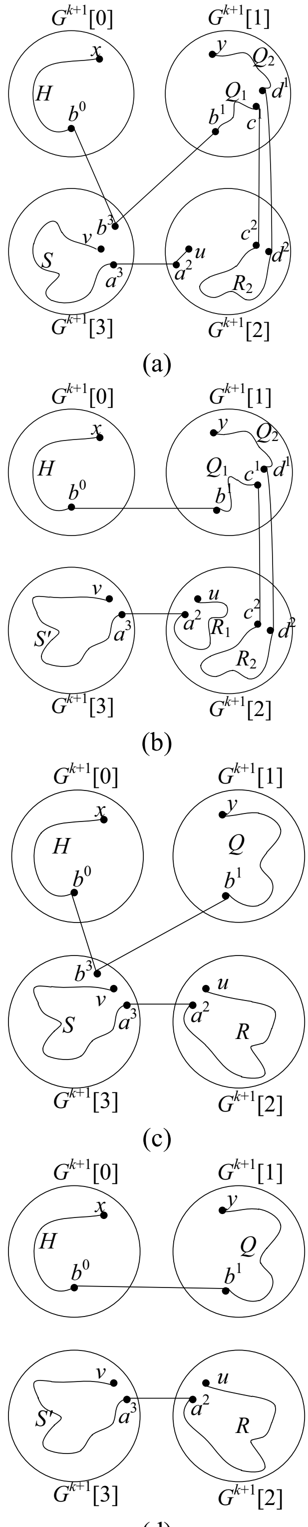
For $l_1 = 2(4^k)-1$, $P_1 = \langle u, R, a^2, a^3, S', v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q, y\rangle$ are the required paths as shown in Figure 6(d).

Case 2. ($\sigma^{k+1}(u) \neq \sigma^{k+1}(v)$ and $|\{\sigma^{k+1}(u), \sigma^{k+1}(v)\} \cap \{\sigma^{k+1}(x), \sigma^{k+1}(y)\}| = 1$): Without loss of generality, assume that $\sigma^{k+1}(u)=\sigma^{k+1}(y)=1$ and $\sigma^{k+1}(v)=2$ such that u and v are included in $G^{k+1}[1]$ and $G^{k+1}[2]$, respectively. Three cases are necessary to be discussed depending on the values of $d_G(u, v)$ and l_1.

Case 2.1. ($d_G(u, v) \leq l_1 \leq 4^k-1$ with $d_G(u, v) = 1$):

For $l_1=1$, we have $P_1 = \langle u, v\rangle$. There is a vertex a^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^1 \notin \{y, u\}$. By Lemma 1, there is a Hamiltonian path H in $G^{k+1}[0]$ joining x and a^0. By Lemma 2, there is a Hamiltonian path Q in $G^{k+1}[1]\backslash\{u\}$ joining a^1 and y. Besides, Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1. By Lemma 2, there exist a Hamiltonian path R in $G^{k+1}[2]\backslash\{v\}$ joining b^2 and c^2. Besides, R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2. By Lemma 1, there is a Hamiltonian path S in

$G^{k+1}[3]$ joining d^3 and e^3. Hence, $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$, and P_1 and P_2 are the required paths.



(d)

Figure 6. Paths P_1 and P_2 constructed by Case 1.3. (a) l_1=4^k. (b) 4^k+1≤l_1≤2(4^k)–3. (c) l_1=2(4^k)–2. (d) l_1=2(4^k)–1.

For l_1=2, we have $P_1 = \langle u, u^3, v\rangle$. By Lemma 2, there is a Hamiltonian path S' in $G^{k+1}[3]\backslash\{u^3\}$ joining d^3 to e^3. Hence, $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S', e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$, and P_1 and P_2 are the required paths.

For $3 \le l_1 \le 4^k-1$, there are two vertices $a^1 \in N_{G^{k+1}[2]}(u)$ and b^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^1 \ne y$ and $b^1 \notin \{y, u, a^1\}$. By Lemma 1, there is a Hamiltonian path H in $G^{k+1}[0]$ joining x and b^0. By the induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining u and a^2 with $l(Q_1)$=1; (2) Q_2 is a path joining b^1 and y with $l(Q_2)$=4^k–3, and (3) $Q_1 \cup Q_2$ spans $G^{k+1}[1]$. By Lemma 4, Q_2 can be written as $\langle b^1, Q_{21}, c^1, d^1, Q_{22}, y\rangle$ for some vertices c^1 and d^1 such that $N_{G^{k+1}[3]}(v) \cap N_{G^{k+1}[3]}(a^2) \ne \{c^2, d^2\}$ and $\{v, a^2\} \cap \{c^2, d^2\} = \emptyset$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining a^2 and v with $l(R_1)$=l_1–2; (2) R_2 is a path joining c^2 and d^2 with $l(R_2)$=4^k–2–$l(R_1)$, and (3) $R_1 \cup R_2$ spans $G^{k+1}[2]$. R_2 can be written as $\langle c^2, R_{21}, e^2, f^2, R_{22}, d^2\rangle$ for some vertices e^2 and f^2. If $l(R_2)$ = 1, then e^2=c^2 and f^2=d^2. By Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining e^3 and f^3. Hence, $P_1 = \langle u, a^1, a^2, R_1, v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q_{21}, c^1, c^2, R_{21}, e^2, e^3, S, f^3, f^2, R_{22}, d^2, d^1, Q_{22}, y\rangle$ are the required paths.

Case 2.2. ($d_G(u, v) \le l_1 \le 4^k-1$ with $d_G(u, v) \ge 2$):

For $d_G(u, v) \le l_1 \le 4^k-2$, there is a vertex a^0 in $G^{k+1}[0]\backslash\{x\}$ such that $a^1 \notin \{y, u\}$. By Lemma 1, there is a Hamiltonian path H in $G^{k+1}[0]$ joining x and a^0. By Lemma 2, there is a Hamiltonian path Q of $G^{k+1}[1]\backslash\{u\}$ joining a^1 and y. By Lemma 4, Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1 such that $N_{G^{k+1}[3]}(v) \cap N_{G^{k+1}[3]}(u^2) \ne \{b^1, c^1\}$ and $\{v, u^2\} \cap \{b^2, c^2\} = \emptyset$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u^2 and v with $l(R_1)$=l_1–1; (2) R_2 is a path joining b^2 and c^2 with $l(R_2)$= 4^k–2–$l(R_1)$, and (3) $R_1 \cup R_2$ spans $G^{k+1}[2]$. Besides, R_2 can be written as $\langle b^2, R_{21}, d^2, e^2, R_{22}, c^2\rangle$ for some vertices d^2 and e^2. If $l(R_2)$=1, then d^2=b^2 and e^2=c^2. By Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining d^3 and e^3. Hence, $P_1 = \langle u, u^2, R_1, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_{21}, d^2, d^3, S, e^3, e^2, R_{22}, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $l_1 = 4^k-1$, there is a Hamiltonian path R in $G^{k+1}[2]\backslash\{c^2\}$ joining u^2 and v by Lemma 2. By Lemma 1, there is a Hamiltonian path S' in $G^{k+1}[3]$ joining b^3 and c^3. Hence, $P_1 = \langle u, u^2, R, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^3,$

S', c^3, c^2, c^1 Q_2, $y\rangle$ are the required paths.

Case 2.3.($4^k \le l_1 \le 2(4^k)-1$):

For $l_1=4^k$, there is $a^1 \in N_{G^{k+1}[1]}(u)-\{y\}$ such that $a^2 \ne v$. Since $|(N_{G^{k+1}[1]}(u) \cap N_{G^{k+1}[1]}(a^1))|=2$, we have $|N_{G^{k+1}[1]}(y)-(N_{G^{k+1}[1]}(u) \cap N_{G^{k+1}[1]}(a^1))| \ge 3k-2 \ge 4$. Thus, there is a vertex $b^1 \in N_{G^{k+1}[1]}(y)-(N_{G^{k+1}[1]}(u) \cap N_{G^{k+1}[1]}(a^1))$ in $G^{k+1}[1]\backslash\{u, a^1\}$ such that $b^0 \ne x$. By Lemma 1, there is a Hamiltonian path H in $G^{k+1}[0]$ joining x and b^0. By the induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining u and a^1 with $l(Q_1)=1$; (2) Q_2 is a path joining b^1 and y with $l(Q_2)=4^k-3$, and (3) $Q_1 \cup Q_2$ spans $G^{k+1}[1]$. Since $l(Q)=4^k-3$, Q can be written as $\langle b^1, Q_1, c^1, d^1, Q_2, y\rangle$ for some vertices c^1 and d^1 such that $c^2 \ne v$. By Lemma 2, there is a Hamiltonian path R in $G^{k+1}[2]\backslash\{d^2\}$ joining a^2 and v. By Lemma 1, there is a Hamiltonian path S of $G^{k+1}[3]$ joining c^3 and d^3. Hence, $P_1 = \langle u, Q_1, a^1, a^2, R, v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q_{21}, c^1, c^3, S, d^3, d^2, d^1, Q_{22}, y\rangle$ are the required paths.

For $4^k+1 \le l_1 \le 2(4^k)-3$, since $N_{G^{k+1}[1]}(u) \cap N_{G^{k+1}[1]}(a^2) \ne \{b^1, y\}$, by the induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining u and a^1 with $l(Q_1)=l_1-4^k$; (2) Q_2 is a path joining b^1 and y with $l(Q_2)=4^k-2-l(Q_1)$, and (3) $Q_1 \cup Q_2$ spans $G^{k+1}[1]$. By Lemma 1, there is a Hamiltonian path R' in $G^{k+1}[2]$ joining a^2 and v. Hence, $P_1 = \langle u, Q_1, a^1, a^2, R', v\rangle$ and $P_2 = \langle x, H, b^0, b^1, Q_{21}, c^1, c^3, S, d^3, d^1, Q_{22}, y\rangle$ are the required paths.

For $l_1 = 2(4^k)-2$, there are two vertices a^1 in $G^{k+1}[1]\backslash\{u, y\}$ and b^0 in $G^{k+1}[0]\backslash\{x\}$ such that $b^1 \notin \{y, a^1\}$. By Lemma 1, there are a Hamiltonian path R' in $G^{k+1}[2]$ joining a^2 and v and a Hamiltonian path S' in $G^{k+1}[3]$ joining b^3 and y^3. By Lemma 2, there is a Hamiltonian path Q in $G^{k+1}[1]\backslash\{y\}$ joining u and a^1. Hence, $P_1 = \langle u, Q, a^1, a^2, R', v\rangle$ and $P_2 = \langle x, H, b^0, b^3, S', y^3, y\rangle$ are the required paths.

For $l_1 = 2(4^k)-1$, there is a Hamiltonian path S'' in $G^{k+1}[3]\backslash\{a^3\}$ joining b^3 and y^3 by Lemma 2. Hence, $P_1 = \langle u, Q, a^1, a^3, a^2, R', v\rangle$ and $P_2 = \langle x, H, b^0, b^3, S'', y^3, y\rangle$ are the required paths.

Case 3. ($\sigma^{k+1}(u) \ne \sigma^{k+1}(v)$ and $|\{\sigma^{k+1}(u), \sigma^{k+1}(v)\} \cap \{\sigma^{k+1}(x), \sigma^{k+1}(y)\}| = 2$): Without loss of generality, assume that $\sigma^{k+1}(u)=\sigma^{k+1}(x)=0$ and $\sigma^{k+1}(v)=\sigma^{k+1}(y)=1$ such that u and v are included $G^{k+1}[0]$ and $G^{k+1}[1]$, respectively. Three cases are necessary to be discussed depending on the values of $d_G(u, v)$ and l_1.

Case 3.1. ($d_G(u, v) \le l_1 \le 4^k-1$ with $d_G(u, v) = 1$):

For $l_1=1$, we have $P_1 = \langle u, v\rangle$. There is a vertex a^0 in $G^{k+1}[0]\backslash\{x, u\}$ such that $a^1 \ne y$. By Lemma 2, there are two Hamiltonian paths H in $G^{k+1}[0]\backslash\{u\}$ and Q in $G^{k+1}[1]\backslash\{v\}$ such that H joins x and a^0, and Q joins a^1 and y. Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1. By Lemma 1, there is a Hamiltonian path R in $G^{k+1}[2]$ joining b^2 and c^2, and R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2. By Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining d^3 and e^3. Hence, $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$, and P_1 and P_2 are the required paths.

For $l_1=2$, by Lemma 2, there is a Hamiltonian path R' in $G^{k+1}[2]\backslash\{u^2\}$ joining b^2 and c^2, and R' can be written as $\langle b^2, R'_1, d^2, e^2, R'_2, c^2\rangle$ for some vertices d^2 and e^2. Hence, $P_1 = \langle u, u^2, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R'_1, d^2, d^3, S, e^3, e^2, R'_2, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $3 \le l_1 \le 4^k-1$, there is a vertex $f^0 \in N_{G^{k+1}[2]}(u)-\{x\}$ such that $f^1 \ne y$. Since $|(N_{G^{k+1}[0]}(u) \cap N_{G^{k+1}[0]}(f^0))|=2$, we have $|N_{G^{k+1}[0]}(x)-(N_{G^{k+1}[0]}(u) \cap N_{G^{k+1}[0]}(f^0))| \ge 3k-2 \ge 4$. Thus, there is a vertex $g^0 \in N_{G^{k+1}[0]}(y)-(N_{G^{k+1}[0]}(u) \cap N_{G^{k+1}[0]}(f^1))$ in $G^{k+1}[0]\backslash\{u, f^0\}$ such that $g^1 \ne y$. Since $N_{G^{k+1}[0]}(u) \cap N_{G^{k+1}[0]}(a^0) \ne \{x, b^0\}$, by the induction hypothesis, there exist two disjoint paths H_1 and H_2 such that (1) H_1 is a path joining u and f^0 with $l(H_1)=l_1-2$; (2) H_2 is a path joining x and g^0 with $l(H_2)=4^k-2-l(H_1)$, and (3) $H_1 \cup H_2$ spans $G^{k+1}[0]$. Again, by the induction hypothesis, there exist two disjoint paths Q'_1 and Q'_2 such that (1) Q'_1 is a path joining f^1 and v with $l(Q'_1)=1$; (2) Q_2 is a path joining g^1 and y with $l(Q'_2)=4^k-3$, and (3) $Q'_1 \cup Q'_2$ spans $G^{k+1}[1]$. Besides, Q'_2 can be written as $\langle a^1, Q'_{21}, b^1, c^1, Q'_{22}, y\rangle$ for some vertices b^1 and c^1. Hence, $P_1 = \langle u, H_1, f^0, f^1, v\rangle$ and $P_2 = \langle x, H_2, g^0, g^1, Q'_{21}, b^1, b^2, R_1, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q'_{22}, y\rangle$ are the required paths.

Case 3.2. ($d_G(u, v) \le l_1 \le 4^k-1$ with $d_G(u, v) \ge 2$):

For $l_1 = d_G(u, v)$ with $d_G(u, v)=2$, we have $l_1= 2$, which implies $N_G(u) \cap N_G(v) \ne \{x, y\}$. Since $N_G(u) \cap N_G(v) = \{u^1, v^0\}$, we have $\{u^1, v^0\} \ne \{x, y\}$. Without loss of generality, assume that $u^1 \ne y$. There are two vertices $a^1 \notin N_{G^{k+1}[1]}(u^1) \cap N_{G^{k+1}[1]}(v)$ in $G^{k+1}[3]\backslash\{u^1, v, y, x^1\}$, and b^2 in $G^{k+1}[2]\backslash\{a^2\}$. By the induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining u^1 and v with $l(Q_1)=l_1-1$; (2) Q_2 is a path joining a^1 and y with $l(Q_2)=4^k-2-l(Q_1)$, and (3) $Q_1 \cup Q_2$ spans $G^{k+1}[1]$. By Lemma 2, there is a Hamiltonian path H in $G^{k+1}[0]\backslash\{u\}$

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joining x and a^0. By Lemma 1, there are two Hamiltonian paths R in $G^{k+1}[2]$ and S in $G^{k+1}[3]$ such that R joins a^2 and b^2, and S joins a^3 and b^3. Hence, $P_1 = \langle u, u^1, Q_1, v\rangle$ and $P_2 = \langle x, H, a^0, a^3, S, b^3, b^2, R, a^2, a^1, Q_2, y\rangle$ are the required paths.

For $l_1= d_G(u, v)$ with $d_G(u, v) \geq 3$, If $u^1\neq y$ or $v^0\neq x$, then the proof is the same as the situation that $l_1= d_G(u, v)$ with $d_G(u, v)=2$. On the other hand ($u^1=y$ and $v^0=x$), there are four vertices a^0, b^0, c^1, and d^2 such that $a^0 \in N_{G^{k+1}[0]}(u)$ and $d(a^0, v)= d_G(u, v)-1$, $b^0 \notin N_{G^{k+1}[0]}(u)\cap N_{G^{k+1}[0]}(a^0)$ is included in $G^{k+1}[0]\setminus \{u, a^0, x\}$, $c^1 \notin N_{G^{k+1}[1]}(a^1)\cap N_{G^{k+1}[1]}(v)$ is included in $G^{k+1}[1]\setminus\{v, a^0, y\}$, and d^2 is included in $G^{k+1}[2]\setminus\{c^2\}$ and $d^3\neq b^3$. Since $d(a^1, v)=d(a^0, v)-1$, we have $d(a^1, v)= d_G(u, v)-2$. By the induction hypothesis, there exist two disjoint paths H_1 and H_2 such that (1) H_1 is a path joining u and a^0 with $l(H_1)=1$; (2) H_2 is a path joining x and b^0 with $l(H_2)=4^k-3$, and (3) $H_1\cup H_2$ spans $G^{k+1}[0]$. Again, by the induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining a^1 and v with $l(Q_1)=l_1-2$; (2) Q_2 is a path joining c^1 and y with $l(Q_2)=4^k-2-l(Q_1)$, and (3) $Q_1\cup Q_2$ spans $G^{k+1}[1]$. By Lemma 1, there are two Hamiltonian paths R in $G^{k+1}[2]$ and S in $G^{k+1}[3]$ such that R joins c^2 and d^2, and S joins b^3 and d^3. Hence, $P_1 = \langle u, H_1, a^0, a^1, Q_1, v\rangle$ and $P_2 = \langle x, H_2, b^0, b^3, S, d^3, d^2, R, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $d_G(u, v)+1 \leq l_1 \leq 4^k-1$, there is a vertex a^0 in $G^{k+1}[0]\setminus\{x, u\}$ such that $a^1\neq y$. By Lemma 2, there are two Hamiltonian paths H in $G^{k+1}[0]\setminus\{u\}$ and Q in $G^{k+1}[1]\setminus\{v\}$ such that H joins x and a^0, and Q joins a^1 and y. By Lemma 4, Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1 such that $N_{G^{k+1}[3]}(u^2)\cap N_{G^{k+1}[3]}(v^2) \neq \{b^2, c^2\}$ and $\{u, v^2\}\cap\{b^2, c^2\} = \emptyset$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u^2 and v^2 with $l(R_1)=l_1-2$; (2) R_2 is a path joining b^2 and c^2 with $l(R_2)=4^k-2-l(R_1)$, and (3) $R_1\cup R_2$ spans $G^{k+1}[2]$. Besides, R_2 can be written as $\langle b^2, R_{21}, d^2, e^2, R_{22}, c^2\rangle$ for some vertices d^2 and e^2. By Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining d^3 and e^3. Hence, $P_1 = \langle u, u^2, R_1, v^2, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_{21}, d^2, d^3, S, e^3, e^2, R_{22}, c^2, c^1, Q_2, y\rangle$ are the required paths.

Case 3.3. ($4^k \leq l_1 \leq 2(4^k)-1$):

For $l_1=4^k$, there are two vertices $a^1 \in N_{G^{k+1}[1]}(v)-\{y\}$ and b^1 in $G^{k+1}[1]\setminus\{v, a^1, y\}$ such that $a^0\notin\{u, x\}$ and $b^0\neq x$. By Lemma 2, there is a Hamiltonian path H in $G^{k+1}[0]\setminus\{x\}$ joining u and a^0. By induction hypothesis, there exist two disjoint paths Q_1 and Q_2 such that (1) Q_1 is a path joining a^1 and v with $l(Q_1)=1$; (2) Q_2 is a path joining b^1 to y with $l(Q_2)=4^k-3$, and (3) $Q_1\cup Q_2$ spans $G^{k+1}[1]$. Besides, Q_2 can be written as $\langle b^1, Q_{21}, c^1, d^1, Q_{22}, y\rangle$. By Lemma 1, there are two Hamiltonian paths R in $G^{k+1}[2]$ and S in $G^{k+1}[3]$ such that R joins c^2 and d^2, and S joins x^3 and b^3. Hence, $P_1 = \langle u, H, a^0, a^1, Q_1, v\rangle$ and $P_2 = \langle x, x^3, S, b^3, b^1, Q_{21}, c^1, c^2, R, d^2, d^1, Q_{22}, y\rangle$ are the required paths.

For $4^k+1 \leq l_1 \leq 2(4^k)-3$, there exists a Hamiltonian path Q' in $G^{k+1}[1]\setminus\{v\}$ joining b^1 and y by Lemma 2. By Lemma 4, Q' can be written as $\langle b^1, Q_1', b^1, c^1, Q_2', y\rangle$ for some vertices b^1 and c^1 such that $N_{G^{k+1}[3]}(a^3)\cap N_{G^{k+1}[3]}(v^3) \neq \{b^3, c^3\}$ and $\{a^3, v^3\}\cap\{b^3, c^3\} = \emptyset$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u and v with $l(R_1)=l_1-4^k$; (2) R_2 is a path joining c^2 to d^2 with $l(R_2)= 4^k-2-l(R_1)$, and (3) $R_1\cup R_2$ spans $G^{k+1}[2]$. Hence, $P_1 = \langle u, H, a^0, a^2, R_{1,}, v^3, v\rangle$ and $P_2 = \langle x, x^3, S, b^3, b^1, Q_1', c^1, c^2, R_2, d^2, d^1, Q_2', y\rangle$ are the required paths.

For $l_1 = 2(4^k)-2$, there is a Hamiltonian path R' of $G^{k+1}[2]\setminus\{b^2\}$ joining a^2 and v^2 by Lemma 2. Hence, $P_1 = \langle u, H, a^0, a^2, R', v^3, v\rangle$ and $P_2 = \langle x, x^3, S, b^3, b^2, b^1, Q', y\rangle$ are the required paths.

For $l_1 = 2(4^k)-1$, there is a Hamiltonian path R'' in $G^{k+1}[2]$ joining a^2 and v^2 by Lemma 1. Hence, $P_1 = \langle u, H, a^0, a^2, R'',, v^3, v\rangle$ and $P_2 = \langle x, x^3, S, b^3, b^1, Q', y\rangle$ are the required paths.

Case 4. ($\sigma^{k+1}(u)=\sigma^{k+1}(v)$ and $\{\sigma^{k+1}(u)\}\cap\{\sigma^{k+1}(x), \sigma^{k+1}(y)\} = \emptyset$): Without loss of generality, assume that $\sigma^{k+1}(u)=\sigma^{k+1}(v)=2$ such that u and v are both included in $G^{k+1}[2]$.

For $d_G(u, v) \leq l_1 \leq 4^k-3$, there is a vertex a^0 in $G^{k+1}[0]\setminus\{x\}$ such that $a^1 \neq y$ and $a^3\notin\{u, v\}$. By Lemma 1, there is a Hamiltonian path H in $G^{k+1}[0]$ joining x and a^0, and H can be written as $\langle x, H_1, b^1, c^1, H_2, a^0\rangle$ for some vertices b^1 and c^1. By Lemma 1, there is a Hamiltonian path Q in $G^{k+1}[1]$ joining a^1 and y. By Lemma 4, Q can be written as $\langle a^1, Q_1, d^1, e^1, Q_2, y\rangle$ for some vertices d^1 and e^1 such that $N_{G^{k+1}[3]}(u)\cap N_{G^{k+1}[3]}(v) \neq \{d^2, e^2\}$ and $\{u, v\}\cap\{d^2, e^2\} = \emptyset$. By the induction hypothesis, there exist two disjoint paths R_1 and R_2 such that (1) R_1 is a path joining u and v with $l(R_1)=l_1$; (2) R_2 is a path joining c^2 and d^2 with $l(R_2)=4^k-2-l(R_1)$, and (3) $R_1\cup R_2$ spans $G^{k+1}[2]$. By Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining b^3 and c^3. Hence, $P_1 = \langle u, R_1, v\rangle$ and $P_2 = \langle x, H_1, b^0, b^3, S, c^3, c^0, H_2, a^0, a^1, Q_1, d^1, d^2, R_2, e^2, e^1,$

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$Q_2, y\rangle$ are required paths.

For $l_1=4^k-2$, there is a Hamiltonian path R in $G^{k+1}[2]\backslash\{a^2\}$ joining u and v by Lemma 2. Hence, $P_1 = \langle u, R, v\rangle$ and $P_2 = \langle x, H_1, b^0, b^3, S, c^3, c^0, H_2, a^0, a^2, a^1, Q, y\rangle$ are the required paths.

For $l_1=4^k-1$, there is a Hamiltonian path R' in $G^{k+1}[2]$ joining u and v by Lemma 1. Hence, $P_1 = \langle u, R', v\rangle$ and $P_2 = \langle x, H_1, b^0, b^3, S, c^3, c^0, H_2, a^0, a^1, Q, y\rangle$ are the required paths.

For $4^k \le l_1 \le 2(4^k)-4$, obviously, path R can be written as $\langle u, R_1, f^2, g^2, R_2, v\rangle$ for some vertices f^2 and g^2. By Lemma 4, H can be written as $\langle u, H_1, b^0, c^0, H_2, a^0\rangle$ for some vertices b^0 and c^0 such that $N_{G^{k+1}[3]}(e^3)\cap N_{G^{k+1}[3]}(f^3) \neq \{b^0, c^0\}$ and $\{f^3, g^3\}\cap\{b^3, c^3\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths S_1 and S_2 such that (1) S_1 is a path joining f^3 to g^3 with $l(S_1)=l_1-4^k+1$; (2) S_2 is a path joining b^3 to c^3 with $l(S_2)=4^k-2-l(S_1)$, and (3) $S_1\cup S_2$ spans $G^{k+1}[3]$. Hence, $P_1 = \langle u, R_1, f^2, f^3, S_1, g^3, g^2, R_2, v\rangle$ and $P_2 = \langle x, H_1, b^0, b^3, S_2, c^3, c^0, H_2, a^0, a^2, a^1, Q, y\rangle$ are the required paths.

For $l_1=2(4^k)-3$, there is a Hamiltonian path S' in $G^{k+1}[3]\backslash\{a^3\}$ joining f^3 and g^3 by Lemma 2. Hence, $P_1 = \langle u, R_1, f^2, f^3, S', g^3, g^2, R_2, v\rangle$ and $P_2 = \langle x, H, a^0, a^3, a^2, a^1, Q, y\rangle$ are the required paths.

For $l_1=2(4^k)-2$, there exist a Hamiltonian path S'' in $G^{k+1}[3]$ joining f^3 and g^3 by Lemma 1. Hence, $P_1 = \langle u, R_1, f^2, f^3, S'', g^3, g^2, R_2, v\rangle$ and $P_2 = \langle x, H, a^0, a^2, a^1, Q, y\rangle$ are the required paths.

For $l_1=2(4^k)-1$, obviously, path R' can be written as $\langle u, R'_1, f^2, g^2, R'_2, v\rangle$ for some vertices f^2 and g^2. Hence, $P_1 = \langle u, R'_1, f^2, f^3, S, g^3, g^2, R'_2, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q, y\rangle$ are the required paths.

Case 5. ($\sigma^{k+1}(u)=\sigma^{k+1}(v)$ and $|\{\sigma^{k+1}(u)\}\cap\{\sigma^{k+1}(x), \sigma^{k+1}(y)\}| = 1$): Without loss of generality, assume that $\sigma^{k+1}(u)=\sigma^{k+1}(v)=0$ such that u and v are both included in $G^{k+1}[0]$.

For $d_G(u, v) \le l_1 \le 4^k-3$. Since $|N_{G^{k+1}[0]}(x) - (N_{G^{k+1}[0]}(u)\cap N_{G^{k+1}[1]}(v))| \ge 4$, there is a vertex $a^0 \in N_{G^{k+1}[1]}(x) - (N_{G^{k+1}[1]}(u)\cap N_{G^{k+1}[1]}(a^1))$ in $G^{k+1}[0]\backslash\{u, v\}$ such that $a^1\neq y$ and $N_{G^{k+1}[0]}(u)\cap N_{G^{k+1}[0]}(v) \neq \{x, a^0\}$. By the induction hypothesis, there exist two disjoint paths H_1 and H_2 such that (1) H_1 is a path joining u to v with $l(H_1)=l_1$; (2) H_2 is a path joining x to a^0 with $l(H_2)=4^k-2-l(H_1)$, and (3) $H_1\cup H_2$ spans $G^{k+1}[0]$. By Lemma 1, there is a Hamiltonian path Q in $G^{k+1}[1]$ joining a^1 and y, and Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1. By Lemma 1, there is a Hamiltonian path R in $G^{k+1}[2]$ joining b^2 and c^2, and R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2. Again, by Lemma 1, there is a Hamiltonian path S in $G^{k+1}[3]$ joining d^1 and e^1. Hence, $P_1 = \langle u, H_1, v\rangle$ and $P_2 = \langle x, H_2, a^0, a^1, Q_1, b^1, b^2, R_1\, d^2, d^3, S, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $l_1=4^k-2$, there is a Hamiltonian path H in $G^{k+1}[0]\backslash\{x\}$ joining u to v by Lemma 2. By Lemma 1, there exists a Hamiltonian path S' in $G^{k+1}[3]$ joining x^3 and a^3. Hence, $P_1 = \langle u, H, v\rangle$ and $P_2 = \langle x, x^3, S', a^3, a^1, Q_1, b^1, b^2, R, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $4^k-1 \le l_1 \le 2(4^k)-5$, $l(H_1)$ and $l(H_2)$ can be reset as 4^k-3 and 1, and H_1 can be written as $\langle u, H_{11}, f^0, g^0, H_{12}, a^0\rangle$ for some vertices f^0 and g^0. By Lemma 4, R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2 such that $N_{G^{k+1}[3]}(f^3)\cap N_{G^{k+1}[3]}(g^3) \neq \{d^3, e^3\}$ and $\{f^3, g^3\}\cap\{d^3, e^3\} = \varnothing$. By the induction hypothesis, there exist two disjoint paths S_1 and S_2 such that (1) S_1 is a path joining f^3 and g^3 with $l(S_1)=l_1-4^k+2$; (2) S_2 is a path joining d^3 and e^3 with $l(S_2)=4^k-2-l(S_1)$, and (3) $S_1\cup S_2$ spans $G^{k+1}[3]$. Hence, $P_1 = \langle u, H_{11}, f^0, f^3, S_1, g^3, g^0, H_{12}, v\rangle$ and $P_2 = \langle x, H_2, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S_2, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $l_1=2(4^k)-4$, there is a Hamiltonian path S'' in $G^{k+1}[3]\backslash\{a^3\}$ joining f^3 and g^3 by Lemma 2. Hence, $P_1 = \langle u, H_{11}, f^0, f^3, S'', g^3, g^0, H_{12}, v\rangle$ and $P_2 =\langle x, H_2, a^0, a^3, a^1, Q_1, b^1, b^2, R, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $l_1=2(4^k)-3$, there is a Hamiltonian path S''' in $G^{k+1}[3]$ joining f^3 to g^3 by Lemma 1. Hence, $P_1 = \langle u, H_{11}, f^0, f^3, S''', g^3, g^0, H_{12}, v\rangle$ and $P_2 =\langle x, H_2, a^0, a^1, Q_1, b^1, b^2, R, c^2, c^1, Q_2, y\rangle$ are the required paths.

For $l_1=2(4^k)-2$, Since $l(H) = 4^k-2$, path H can be written as $\langle u, H_1, f^0, g^0, H_2, v\rangle$ for some vertices f^0 and g^0 such that $g^2\neq a^2$. By Lemma 1, there is a Hamiltonian path R' in $G^{k+1}[2]$ joining x and a^2. Hence, $P_1 = \langle u, H_1, f^0, f^3, S''', g^3, g^0, H_2, v\rangle$ and $P_2 =\langle x, x^2, R', a^2, a^1, Q, y\rangle$ are the required paths.

For $l_1=2(4^k)-1$, there is a Hamiltonian path R'' in $G^{k+1}[2]\backslash\{g^2\}$ joining x^2 and a^2 by Lemma 2. Hence, $P_1 = \langle u, H_1, f^0, f^3, S''', g^3, g^2, g^0, H_2, v\rangle$ and $P_2 =\langle x, x^2, R'', a^2, a^1, Q, y\rangle$ are the required paths. ■

Lemma 6. If a $G(4, m_{r-1}, \ldots, m_1)$ satisfies the $2RP$ property, then a $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies the $2RP$ property, where $r\ge2$ and $m_i\ge4$ for all $1\le i\le r$.

Proof. This lemma is proved by induction on m_r. Obviously, the lemma holds for $m_r=4$. Suppose that $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies the $2RP$ property for all $4\le m_r\le k$. In the following, we prove that $G(k+1, m_{r-1}, \ldots, m_1)$ satisfies the $2RP$ property. Let G denote $G(k+1, m_{r-1}, \ldots, m_1)$ for brevity. Without loss of generality, assume that $l_1\le l_2$ and thus $l_1 \le ((k+1)\times N^r-2)/2$. Moreover, $l_1 \le$

$(k\times N^r-2)-\max\{k, d_G(x, y)\}$, as explained below. If $k \ge d_G(x, y)$, then $l_1 \le (k\times N^r-2)-k$ because $((k+1)\times N^r-2)/2 \le (k\times N^r-2)-k$ can be assured by $k\ge 4$, $N^r\ge 4^r$, and $r\ge 2$. On the other hand ($k < d_G(x, y)$), we have $l_1 \le (k\times N^r-2)-d_G(x, y)$ because $((k+1)\times N^r-2)/2 \le (k\times N^r-2)-d_G(x, y)$ can be assured by $k\ge 4$, $N^r\ge 4^r$, $d_G(x, y)\le r$, and $r\ge 2$.

Let u, v, x and y be four distinct vertices in G. Since $G^r[0]$, $G^r[1]$, …, $G^r[k]$ are vertex-disjoint to each other, there exists a $G^r[j]$ such that none of u, v, x and y is included in $G^r[j]$, where $0\le j\le k$. Without loss of generality, assume that $j=k$.

Since $d_G(u, v) \le l_1$, $d_G(u, v) \le l_1 \le (k\times N^r-2)-\max\{k, d_G(x, y)\}$. The following discussions first exclude the situation that $l_1=2$ with $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) = \{x, y\}$ and $N_G(u)\cap N_G(v) \ne \{x, y\}$. Since $G-G^r[k] \cong G(k, m_{r-1}, \ldots, m_1)$, by the induction hypothesis, there exist two disjoint paths H_1 and H_2 such that (1) H_1 is a path joining u and v with $l(H_1) = l_1$; (2) H_2 is a path joining x and y with $l(H_2) = (k\times N^r-2)-l_1$, (3) $H_1\cup H_2$ spans $G-G^r[k]$. Since $l(H_2)\ge k$, by Lemma 3, there exists an m-edge in H_2 where $m\ne r$. Thus, H_2 can be written as $\langle x, H_{21}, a, b, H_{22}, y\rangle$ for some vertices a and b such that a and b is connected by m-edge and $a^k\ne b^k$. By Lemma 1, there is a Hamiltonian path Q in $G^r[k]$ joining a^k and b^k. Hence $P_1 = \langle u, H_1, v\rangle$ and $P_2 = \langle x, H_{21}, a, a^k, Q, b^k, b, H_{22}, y\rangle$ are the required paths as shown in Figure 7(a).

The rest of this proof considers the situation that $l_1=2$ with $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) = \{x, y\}$ and $N_G(u)\cap N_G(v) \ne \{x, y\}$. Since $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) \subset N_G(u)\cap N_G(v)$ and u (v) is not in $G^r[k]$, there is a vertex $t^k \in N_G(u)\cap N_G(v)$ in $G^r[k]$, such that u (v) $\in N_G(t^k)$; u (v) and t^k are r-neighbors, and u and v are r-neighbors. Therefore, u^k (v^k) $= t^k$ and $d_G(u, v)=1$. Since $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) = \{x, y\}$ and u and v are r-neighbors, only four vertices u, v, x and y are connected by r-edge in $G-G^r[k]$ such that $\sigma^r(u)\ne\sigma^r(v)\ne\sigma^r(x)\ne\sigma^r(y)$ and $k=4$. Without loss of generality, assume that $\sigma^r(x)=0$, $\sigma^r(y)=1$, $\sigma^r(u)=2$, and $\sigma^r(v)=3$, such that u, v, x, and y are included in $G^r[0]$, $G^r[1]$, $G^r[2]$, and $G^r[3]$, respectively. There is a vertex a^0 in $G^r[0]\backslash\{x\}$ such that $a^1\ne y$. By Lemma 1, there are two Hamiltonian paths H in $G^r[0]$ and Q in $G^r[1]$ such that H joins x and a^0, and Q joins a^1 and y. Since $l(Q) \ge 4^{r-1}-1$, Q can be written as $\langle a^1, Q_1, b^1, c^1, Q_2, y\rangle$ for some vertices b^1 and c^1 such that $\{b^2, c^2\}\cap\{u\} = \varnothing$. By Lemma 2, there is a Hamiltonian path R in $G^r[2]\backslash\{u\}$ joining b^2 and c^2, and R can be written as $\langle b^2, R_1, d^2, e^2, R_2, c^2\rangle$ for some vertices d^2 and e^2. By Lemma 2, there is a Hamiltonian path S in $G^r[3]\backslash\{v\}$ joining d^3 and e^3, and S can be written as $\langle d^3, S_1, f^3, g^3, S_2, e^3\rangle$ for some vertices f^3 and g^3. Again, by Lemma 2, there is a Hamiltonian path T in $G^r[4]\backslash\{u^4\}$ joining f^4 and g^4. Hence, $P_1 = \langle u, t^4, v\rangle$ and $P_2 = \langle x, H, a^0, a^1, Q_1, b^1, b^2, R_1, d^2, d^3, S_1, f^3, f^4, T, g^4, g^3, S_2, e^3, e^2, R_2, c^2, c^1, Q_2, y\rangle$ are the required paths as shown in Figure 7(b).

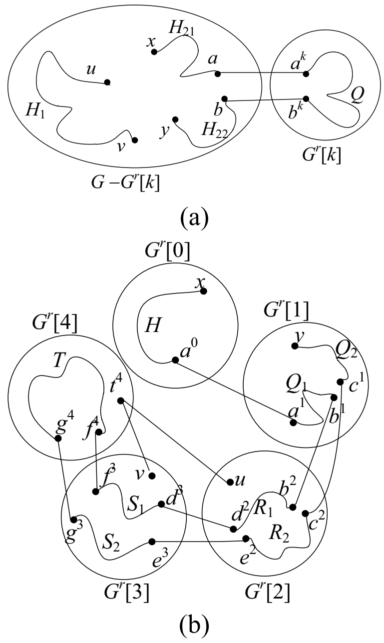

Figure 7. Paths P_1 and P_2 constructed by Lemma 6, when (a) $d_G(u, v) \le l_1 \le (k\times N^r-2)-\max\{k, d_G(x, y)\}$ except that $l_1=2$ with $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) = \{x, y\}$ and $N_G(u)\cap N_G(v) \ne \{x, y\}$. (b) $l_1=2$ with $N_{G-G^r[k]}(u)\cap N_{G-G^r[k]}(v) = \{x, y\}$ and $N_G(u)\cap N_G(v) \ne \{x, y\}$.

Theorem 1. A $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies 2RP-property, where $m_i\ge 4$ for all $1\le i\le r$.

Proof. If $r=1$, then $G(m_1)$ is a complete graph of m_1 vertices and thus this theorem holds. Let $4^{(x)}$ represent 4, 4, …, 4 of length x. On the other hand ($r\ge 2$), by Lemma 5, $G(4^{(r)})$ satisfies 2RP-property. According to Lemma 6, $G(m_r, 4^{(r-1)})$ satisfies 2RP-property. Since $G(m_r, 4^{(r-1)}) \cong G(4^{(r-1)}, m_r)$, $G(4^{(r-1)}, m_1) = G(4^{(r-1)}, m_r) \cong G(m_r, 4^{(r-1)})$ satisfies 2RP-property, where $m_1=m_r\ge 4$. Again, by Lemma 6, $G(m_r, 4^{(r-2)}, m_1)$ satisfies 2RP-property, where $m_r\ge 4$ and $m_1\ge 4$. Since $G(m_r, 4^{(r-2)}, m_1) \cong G(4^{(r-2)}, m_r, m_1)$, $G(4^{(r-2)}, m_2, m_1) = G(4^{(r-2)}, m_r, m_1) \cong G(m_r, 4^{(r-2)}, m_1)$ satisfies 2RP-property, where $m_2=m_r\ge 4$. By repeating the deduction above r times, we have that $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies 2RP-property, where $m_i\ge 4$ for all $1\le i\le r$. ■

4 Conclusion

The 2RP-property of an interconnection network indicates the path embedding capability of the network. This work first demonstrates that a $G(m_r, m_{r-1}, \ldots, m_1)$ is 1-Hamiltonian-connected, where $m_i\ge 3$ for all $1\le i\le r$. Then, by the aids of this

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property, our study shows that a $G(m_r, m_{r-1}, \ldots, m_1)$ satisfies 2RP-property, where $m_i \geq 4$ for all $1 \leq i \leq r$.

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